

Q Loss Function vs Cost Function

The simplest explanation you'll ever read



Loss Function

Cost Function



Why do ML beginners get confused here?

- Terminology sounds similar
- Both relate to model error
- Used in training





Loss Function = Error for ONE sample

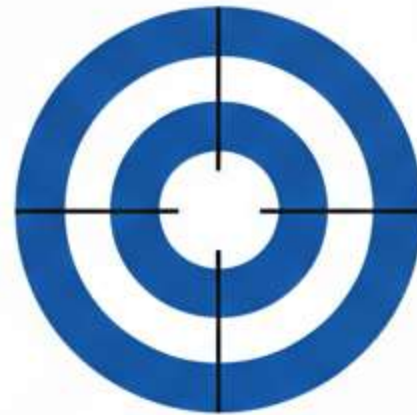
Measures how wrong the model is for a single prediction.

Mini Example

Actual = 10, Pred = 6 $\rightarrow$ Loss = 4



Error = 4 $\rightarrow$



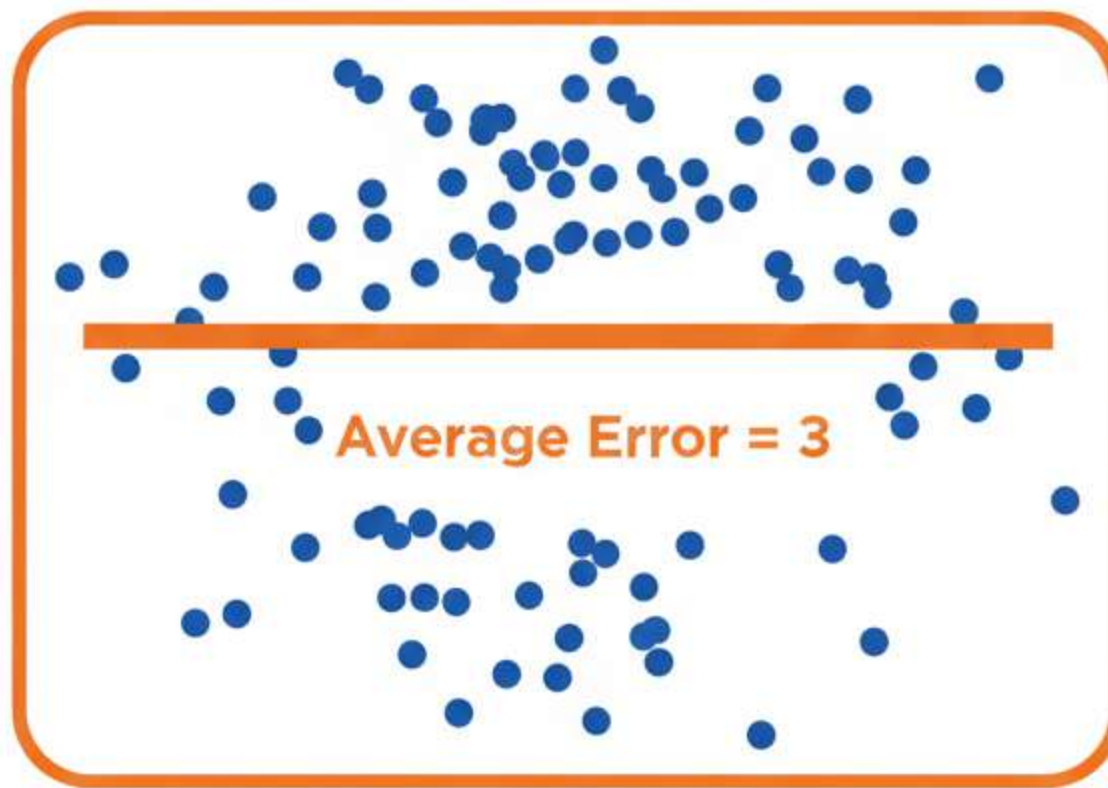


Cost Function = Average error for ALL samples

Measures how wrongl performance at for a single
the dataset.

Mini Example:

Avg loss = 3



The Car Analogy



- **Loss = Speed at a moment
(speedometer)**

(Your instantaneous speed)

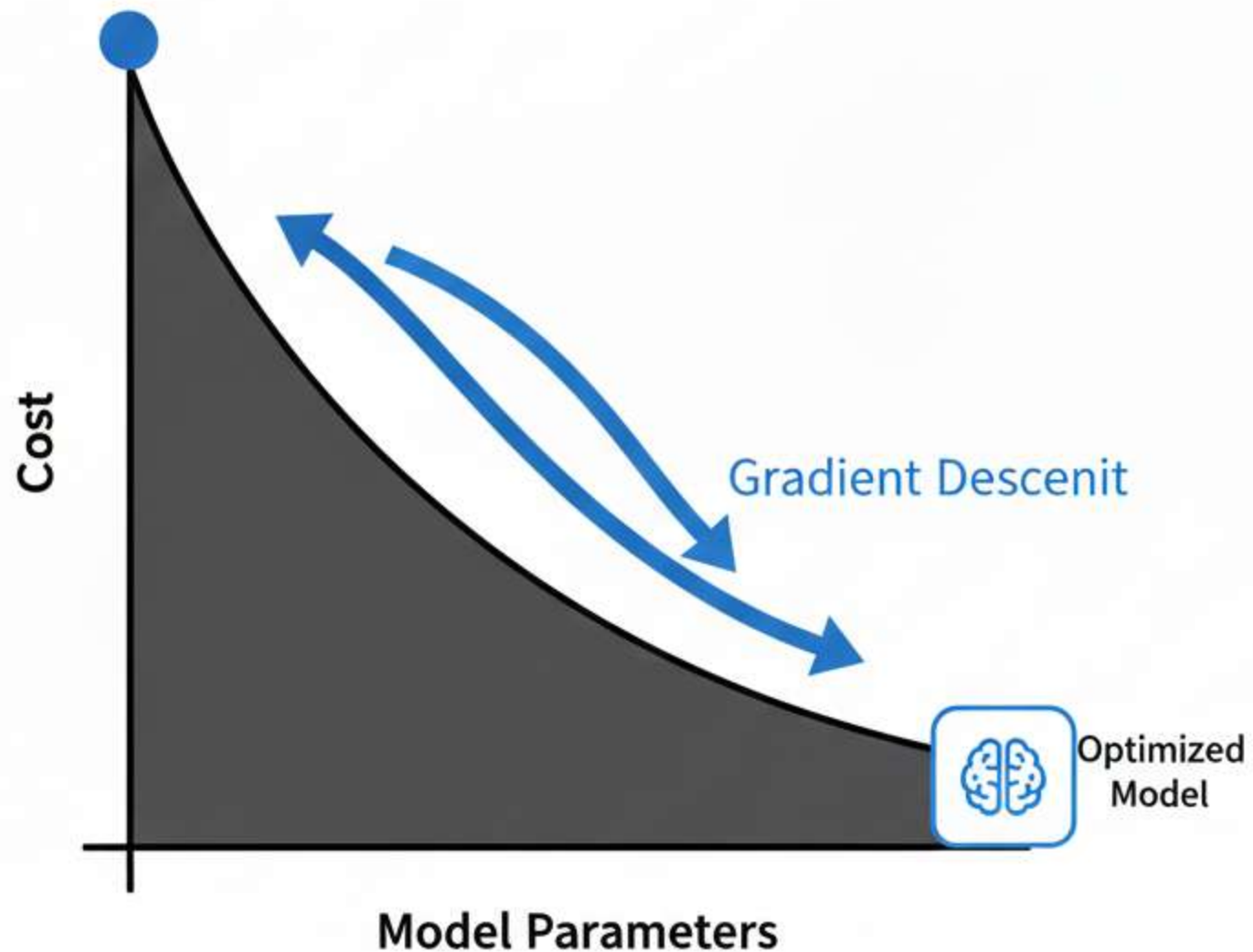


- **Cost = Average speed
of the journey**

(Average speed over the trip)

How models learn using loss & cost

- Loss is computed for each sample
- Cost = average of all losses
- Gradient Descent minimizes the cost





TL:DR

● LOSS

- Error per sample

● COST

- Average error across dataset

Optimize cost, not individual loss

